WEBASTRA-PROBLEM STATEMENTS

SENIOR CATEGORY: (open for ALL)

Problem Statement 1: Mentor Connect

Mentoring during career and education is a vital aspect for the success of a candidate, bringing amazingly positive changes in the life of a mentee. Today, India's demographic advantage has led to infinite career directions. However, emerging career paths require candidates to be coached and mentored for better progress.

This platform can be developed by students who recognize the importance of knowledge sharing and aim to bring mentors and mentees together. The platform should enable candidates to connect with mentors based on their availability, allowing mentors to assist with career guidance, skill development, and industry-related concerns. Mentors could be senior industry leaders or subject matter experts with deep knowledge of job intricacies in their respective fields.

This platform will help candidates connect with mentors across industries, improving their employment opportunities through expert guidance and referrals including features like Automated Calendar Booking, System Embedded Video Call Feature.

Problem Statement 2: Web Application for Direct Market Access for Farmers (contract-based farming)

Farmers often face challenges in accessing markets, leading to lower income due to middlemen. This gap restricts their ability to sell produce at fair prices. Description: Create a mobile application that connects farmers directly with consumers and retailers. The application should include features for listing produce, negotiating prices, and managing transactions, thereby reducing dependence on intermediaries. Expected Solution: A user-friendly platform that enables farmers to showcase their products and connect with buyers directly, enhancing their income potential.

Problem Statement 3: Development of web-based Operation & Maintenance tool for asset & consumables inventories and finance management in context of drinking water supply scheme.

Develop a web application to support the operation and maintenance (O&M) of rural drinking water supply schemes, enabling Gram Panchayats (GP) and Public Health Engineering Departments (PHED) to efficiently manage assets, inventories, and financial tasks.

Key Features:
a) **GIS-based Asset Management:** Record and track water supply infrastructure (pumps, pipelines, valves, treatment plants) with attributes like location, installation date, and historical data.
b) **Inventory Management:** Monitor consumables (chemicals, filters, spare parts), forecast demand, and plan replenishments.
c) **Financial Management:** Record receipts from various sources and expenditures for repairs, replacements, and consumables.
d) **Consumer Management & Billing:** Maintain consumer lists, generate bills, and prepare cash books.
e) **Payment Integration:** Provide a payment interface supporting UPI, net banking, and credit cards.
f) **User-Friendly Interface:** Ensure ease of use for GP-level operations with local language support.

Problem Statement 4: Intelligent platform to Interconnect Alumni and Student for Technical Education Department, GMRIT.

This project aims to build a comprehensive Alumni-Student Interaction Platform for GMR Institute of Technology (GMRIT) to foster engagement, mentorship, and collaboration between alumni and current students. The platform should address key challenges such as lack of centralized alumni information, limited structured interaction, and the need for career guidance.

Problem Statement 5: An Interactive Job and Internship Platform for Technical Education Department

In today's competitive job market, graduates face significant challenges in transitioning from education to employment. Most existing platforms fail to provide comprehensive access to job opportunities across the private sector, government sector, and international employment avenues. Additionally, there is a lack of career guidance, counselling, and internship support, making it difficult for students to secure the right opportunities. By developing this integrated career platform, students will gain better connectivity with potential employers, broaden their career opportunities, and receive the necessary support to succeed in today’s job market.

Problem Statement 6: Learning path dashboard for enhancing skills

Instructor shall have educational resources files in different formats like pdf, word, etc. and hyper-links of relevant academic literature Expected solutions: The desired solution is required to: -Learning dashboard showing different reading statistics like reading time of a particular topic. Total finishing time of a particular skill. -The instructor should be able to easily create a learning path along with incorporating learning resources, as mentioned ""inputs."" including videos. -Progress made by the learner should be continuously updated. -The desired solution must follow UX principles

Problem Statement 7: Learning platform for Deaf and Mute and sign language-English/Telugu converter

Develop a web-based learning platform to support deaf and mute students in learning through Indian Sign Language (ISL), enhancing their access to digital educational resources beyond the classroom. The platform should include Telugu to English and English to Telugu Language Interpretation, providing interactive modules for learning and writing practice. It should feature Word & Sentence Learning, teaching basic words with writing exercises. A Mathematics Module should cover tables, basic arithmetic operations, and assessments, while a Science Module should explain concepts through sign language tutorials and writing exercises. The system must support Telugu to Sign Language and English to Sign Language Conversion, translating text/news into sign language, along with Speech-to-Sign & Sign-to-Speech Conversion for real-time communication. Data Analytics & Assessment should generate student progress reports in Mathematics and Science. The platform should accommodate multiple user roles, including teachers, students, parents, and HR, ensuring inclusivity in learning and hiring. Interactive Learning Tools like writing pads, quizzes, and assessments will reinforce learning. Additionally, Inclusivity Features should aid parents unfamiliar with sign language and assist HR in inclusive hiring. This platform will bridge the communication gap, provide a structured learning experience, and promote digital accessibility for specially-abled students in Telugu Sign Language (TSL).

Problem Statement 8: Cargo - Tracking of all packages from arrival to departure in cost effective manner

The cargo tracking system aims to address the challenges of accurately monitoring packages from arrival to departure in a cost-effective manner. Traditional tracking methods often lack real-time visibility, leading to delays, misplaced shipments, and increased operational costs. This web application will provide a comprehensive solution to track package movements, optimize shipping costs, and ensure transparency through automated status updates, real-time monitoring, and efficient resource management. By implementing role-based access, detailed tracking, and automated notifications, the system enhances operational efficiency while reducing errors and improving customer satisfaction.

Problem Statement 9: Development of a Geolocation-Based Attendance Tracking Web Application.

A web application needs to be developed to streamline and automate the attendance tracking of its employees across multiple office locations. This application will leverage geolocation technology to record employee check-in and check-out times along with geo-location as they enter and leave the office premises. Aim of this application is to enhance operational efficiency, reduce manual attendance tracking errors, and provide a seamless experience for employees to manage their work-related movements.

Key Requirements:

1) Geolocation-Based Check-In and Check-Out: Automatically record check-in time and geo-location when an employee enters within a 200-meter radius of his office. Record check-out time when the employee leaves the 200-meter radius. Ensure that each check-in is paired with a corresponding check-out, regardless of the frequency of entries and exits.

2) Manual Location Check-In / Check-Out for Offsite Work: Provide a manual check-in and check-out feature for employees working at locations other than his offices. Automatically suggest relevant locations based on real-time longitude and latitude data, allowing employees to confirm their check-in and check-out at these suggested locations.

3) Calculate the employees total working hours.

4) Data Accuracy and Integrity: Ensure the application maintains accurate and tamper-proof records of all check-in and check-out events. Enable real-time data synchronization and secure storage to prevent data loss and ensure reliability.

Problem statement 10: Drug Inventory and supply chain Tracking system.

With the aim to provide "Right Quantity of "Right Product" on "Right Place" on "Right Time" in "Right Condition" at "Right Cost" for "Right People" and also to streamline the distribution of drugs to institutions and ensure availability of drugs at all times, a new, innovative system named Drug Inventory and supply chain Tracking system is required: To improve efficiency and effectiveness of procurement and distribution systems through robust quality controls To provide dashboard based online monitoring of all activities at each level Tracking of vendor activities like preparation of Supply Order, Shipment etc. Monitoring of Drug consumption pattern at the Hospitals/Medical Institutions Level.

Problem statement 11 : Passenger Experience - Virtual avatar for passenger engagement, details about facilities in the airport and general conversation capabilities

The virtual avatar for passenger engagement aims to enhance the airport experience by providing real-time information about facilities, flight details, and general assistance through an interactive web application. Traditional methods of passenger support often lead to delays and limited access to information. This system addresses these challenges by offering multilingual support, interactive maps for navigation, and real-time flight updates. The avatar engages in general conversation, answers common queries, and provides 24/7 assistance, improving passenger satisfaction and reducing staff workload while ensuring a seamless and efficient airport journey.

JUNIOR CATEGORY: (1ST AND 2ND YEARS)

Problem Statement 1: Farmers Disease Diagnostic/Reporting Portal - web Based

Develop a web-based disease diagnosis and reporting platform to assist farmers in managing livestock and crop health efficiently. The platform should enable farmers to upload images and descriptions of disease symptoms, allowing experts to analyse data, generate suspected disease reports, and provide preventive measures. It should integrate with existing NDLM systems to ensure smooth data submission and real-time alerts to veterinarians for intervention. The system should facilitate timely reporting, enhancing early disease detection and control. Key features include manual diagnosis submission, real-time disease alerts, expert consultation access, data analytics for trend identification, and cost-effective disease management solutions. Additionally, it should support community engagement, allowing farmers to share insights, discuss issues, and adopt sustainable agricultural practices through shared knowledge. The platform will empower farmers, improve farm productivity, and strengthen disease surveillance networks for a more resilient agricultural ecosystem.

Problem Statement 2: Dashboard for Swachhta and LiFE.

Develop a web-based monitoring and data analytics platform to track and institutionalize Swachhta protocols and Lifestyle for Environment (LiFE) practices across the Department of Posts (DoP) network. The platform should enable real-time tracking of Swachhta activities through image-based monitoring, geo-tagging, and timestamped data collection. It should provide centralized data storage, interactive dashboards, and report generation tools to ensure compliance with the Swachhta Action Plan (SAP) 2024. Field units should be able to upload images, log activities, and submit progress reports, while administrators should have access to trend analysis, compliance tracking, and insights for better decision-making. The system should integrate with existing DoP infrastructure, ensuring scalability and ease of use across various locations. Additional features should include customizable templates for reporting, automated notifications, and a feedback mechanism to enhance participation and adherence to Swachhta and LiFE practices. The platform will serve as a centralized governance tool, fostering local ownership, transparency, and standardization in Swachhta implementation across the postal network.

Problem Statement 3: Create a Virtual Herbal Garden that provides an interactive, educational, and immersive experience to users, showcasing the diverse range of medicinal plants used in AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy).

Develop a web-based Virtual Herbal Garden that provides an interactive, educational, and user-friendly experience for exploring medicinal plants used in the AYUSH sector. The platform should feature realistic 3D models of medicinal plants, allowing users to rotate, zoom, and explore each plant from different angles. It should include detailed information such as botanical names, common names, habitat, medicinal uses, and cultivation methods. The platform should integrate multimedia elements like high-quality images, videos, and audio descriptions to enhance the learning experience. Users should be able to search and filter plants based on medicinal use, region, and type. Virtual tours should guide users through curated themes like plants for digestive health, immunity, and skincare. Additional features should allow users to bookmark plants, take notes, and share content on social media. The expected outcome is an engaging, visually appealing, and interactive Virtual Herbal Garden that serves as a comprehensive learning tool for students, practitioners, and enthusiasts, making traditional herbal knowledge more accessible and widely available.

Problem Statement 4: Health Data Information & Management System Web Application (HDIWS)

Develop a web-based platform where hospitals and departments can enter and update data dynamically, enabling real-time monitoring and efficient implementation of Health and Family Welfare Schemes and other health programs. The Super Admin should have access to view and analyse data to provide key inputs for policy formulation and necessary program interventions. The platform should facilitate the seamless flow of health data from facility level to sub-district, district, and state/union territory levels using a Health Data Information & Management System (HDIWS) interface for efficient data management and decision-making.

Problem Statement 5: Development of a web application to provide recreational suitability information of beach locations across India.

Develop a web-based platform to assess and display the suitability of beaches for recreational activities based on real-time oceanic and meteorological conditions. The platform should include location mapping of beaches across India and evaluate key parameters such as ocean alerts (high waves, swell surges, ocean currents, storm surges, tsunamis), wind conditions, and water quality assessments. These parameters will be retrieved using the INCOIS (Indian National Centre for Ocean Information Services) API. The platform should implement a method/algorithm to analyse these parameters and determine the safety and suitability of different coastal locations. Geospatial visualization using color-coded maps should indicate beach suitability, and user-based alert notifications should be provided based on the user's current location if any hazards are detected. This platform will enhance coastal tourism safety by assisting users in planning recreational activities efficiently while mitigating risks.

Problem Statement 6: Online web-based ticketing system

Develop a web-based museum ticket booking system that enables visitors to book tickets online, select time slots, and access exhibition details through an intuitive interface. The system should support multilingual options for a diverse user base. Implement a real-time ticketing and reservation system to prevent double bookings and reduce waiting times. Integrate a secure payment gateway to facilitate online transactions. The platform should include visitor data analytics and reporting tools to help museum administrators manage crowd flow and optimize scheduling. Additionally, features like automated email confirmations, printable/mobile tickets, and event promotions should be incorporated to enhance user convenience.

Problem Statement 7: Develop Effective Career Counselling and Guidance Programs in Schools to Enhance Student Career Choices

Develop a web-based career counselling and guidance platform for schools, enabling students to explore diverse career options and make informed choices. The platform should include career assessment tools, helping students identify strengths and interests through quizzes and aptitude tests. Implement a comprehensive career resource portal with information on various career paths, required skills, educational institutions, and job market trends. Provide virtual mentorship programs, allowing students to connect with industry professionals for guidance. Integrate interactive career exploration tools such as virtual career simulations, videos, and webinars. The platform should support personalized counselling sessions, where students can book one-on-one sessions with career advisors. Enable school administrators to track student progress, generate reports, and organize career-related events. Ensure multi-language support, accessibility features, and mobile responsiveness to enhance usability for all students.

Problem Statement 8: Portal for innovation Excellence Indicators

The web-based Innovation Excellence Tracking Portal aims to streamline the process of tracking, measuring, and showcasing innovation within educational institutions. It will provide secure user authentication with role-based access, integrate data from various sources such as research projects, patents, and funding databases, and offer real-time analytics through interactive dashboards. Key innovation indicators will be customizable to fit departmental needs, with tools for data visualization, automated ranking, recognition mechanisms, and certificate generation. The platform will facilitate collaboration through feedback systems and version control while ensuring compliance with data privacy regulations like GDPR and FERPA. With a user-friendly, responsive design supporting multiple languages, the portal will enhance transparency, encourage participation, and enable data-driven decision-making, ultimately fostering a culture of continuous innovation across institutions.

Problem Statement 9: Freelancing Platform

There is a growing need for a dedicated web-based freelancing platform in India that connects freelancers with short-term and project-based job opportunities. The proposed platform will function as a comprehensive freelance marketplace where employers can post projects, specify requirements, and invite freelancers to apply. Freelancers will have the ability to create detailed profiles showcasing their skills, experiences, and portfolios, along with a rating and review system for credibility. The platform will feature extensive search capabilities, allowing both employers and freelancers to efficiently navigate available opportunities A secure escrow payment system will ensure smooth financial transactions, holding funds until job completion. This platform will empower freelancers by providing them with greater access to work opportunities while enabling businesses to find skilled professionals efficiently. Students are encouraged to explore innovative solutions beyond the outlined scope, tailoring the platform to specific industries or making it more generic for broader applicability.

Problem Statement 10: Learning path dashboard for enhancing skills

The proposed solution is a web-based learning platform that enables instructors to manage and track educational resources efficiently. For a simplified initial implementation, publication records can be provided as a consolidated, bibtex file, but an Excel sheet format is preferred for better usability. The platform will support various educational resource formats, including PDFs, Word documents, hyperlinks, and videos. It will feature an intuitive learning dashboard displaying key statistics, such as reading time for specific topics and total completion time for particular skills. Instructors will have the ability to create structured learning paths, incorporating diverse resources to guide students through their studies. Learners' progress will be continuously tracked and updated in real-time, ensuring a data-driven learning experience. The platform will adhere to UX design principles to provide an engaging and user-friendly experience for both instructors and learners.

Problem Statement 11: Online issuance of Caste and other certificates by Revenue Department need real-time monitoring

The issuance of caste and other certificates by the Revenue Department requires real-time monitoring to assess resource allocation and demand effectively. Currently, resources are allocated without proper analysis, leading to significant backlogs in certain subdivisions where application loads are high. Implementing an efficient monitoring system at the district and central levels will enable data-driven decision-making, ensuring optimal resource distribution. A detailed evaluation of application trends and processing times will facilitate better workforce management, reduce delays, and improve overall service efficiency. This solution aims to enhance transparency, streamline operations, and provide timely issuance of certificates to applicants.

Problem Statement 12: Dynamic route rationalization platform based on real-time traffic and road parameters.

Develop a web-based platform to assist DTC in route rationalization by providing real-time monitoring and management of bus schedules. The system should feature an intuitive dashboard for tracking buses, analyzing route efficiency, and identifying issues such as bunching or delays. It should allow administrators to dynamically update schedules based on live traffic and road conditions, ensuring optimal bus distribution across routes. The platform must support secure user authentication, interactive maps for route visualization, and real-time notifications for better decision-making. By enabling efficient data integration and seamless communication between transport authorities, the system will enhance operational efficiency, reduce congestion, and improve overall public transportation reliability.

Problem Statement 13: Development of a Paperless Scholarship Disbursement System for PMS.

Develop a web-based portal to digitize the submission, verification, and disbursement process of the Post Metric Scholarship, eliminating the need for physical paperwork. The system should feature a user-friendly interface for students to securely upload and manage their documents, along with verify their authenticity. An automated workflow should route verified documents from the SAG Bureau to the Finance Bureau for payment processing, ensuring a seamless transition. Real-time tracking and notification features will keep students updated on their application status, while compliance with data privacy and security standards will safeguard personal information. The platform should be fully digital, enhancing efficiency, reducing processing time, and minimizing the risk of document loss, ultimately streamlining the entire scholarship disbursement process.

Problem Statement 14: Online testing and monitoring of quality of medicines and consumables

Suitable technological module for testing and monitoring of quality of medicines and consumables being received in hospitals would be required so that the system ensures necessary compliance and rejection of low quality supplies without manual intervention

Problem Statement 15: Automated Bus Scheduling and Route Management System for Vizag Transport Corporation

Develop a web-based Automated Bus Scheduling and Route Management System that streamlines bus and crew assignments while optimizing route planning. The system should feature linked and unlinked duty scheduling, allowing administrators to assign specific crews to buses for an entire shift or facilitate handovers between crew members while managing rest periods efficiently. Route management should include mapping existing routes, visually representing the bus network, enabling users to create new routes, and identifying overlaps for better optimization. The platform should provide a user-friendly interface for schedulers, planners, and managers to manage assignments, monitor schedules, and adjust routes dynamically. The system must ensure accessibility across devices, maintain data security, and offer an intuitive experience for seamless operation, enhancing overall efficiency in bus scheduling and route management.

Problem Statement 16: create an Annual Report Portal for institute where all the departmental reports can be integrated and customized

Develop a web-based portal that streamlines the annual report preparation process for educational institutes by focusing on core web development principles. The portal should feature secure user authentication with role-based access control to ensure data privacy. It must support data collection and integration by allowing imports from databases, spreadsheets, and manual entries while ensuring compatibility with existing systems. A key component is data analysis and visualization, offering customizable dashboards, graphs, and charts to present key performance indicators. The system should enable automated report generation in multiple formats such as PDF and HTML, with customizable templates and multimedia support. Collaboration tools should allow multiple users to edit, review, and provide feedback while maintaining version control. The interface should be intuitive, responsive, and accessible across devices, with multilingual support for broader usability. Additionally, the portal must adhere to relevant compliance and security standards to protect institutional data. The final solution should be scalable and adaptable to different educational institutions, ensuring efficiency and ease of use.

Problem Statement 17: Publications summary generator for faculty members profile building

The proposed solution should be able to crawl different popular academic databases, like Google Scholar, DBLP, etc. Often it is desired to have a publication record in a specific period for submission to various accrediting agencies by HEI's; therefore solution may have a provision for such customized queries. Expected solutions: The desired solution is required for generating publication summary for faculty profile showcase. Following are the identified inputs and expected outcomes Input: 1. Faculty names as mentioned in their respective academic profile (research papers) in an excel sheet 2. Year-wise publication record in journals which is exportable in words and excel 3. Year-wise publication record in conferences which is exportable in words and excel 4. Customized publication records in a particular duration

Problem Statement 18: The technological solutions for capturing AQI values through mobile and other forms of stations

DPCC is using different stations at fixed sites for measurement of AQI and other pollution parameters. These fixed stations suffered from various limitations and generally do not give representative values e.g. station located near an industrial area will give higher readings due to proximity to such industrial area which may not be representative of the wider area. Similarly, a temporary construction site/activity near these fixed sites give higher pollution readings due to local reasons. The technological solutions may be required for capturing AQI values through mobile and other forms of stations. Drone would be one of the options where they can record real-time pollution parameters through on-board sensors.

Problem Statement 19: Centralized Automated Solution for Price Estimation & Reasonability.

Develop a web-based procurement price benchmarking system to assist teams in estimating the approximate cost of products and services using publicly available data. The platform should feature secure user registration and login, ensuring controlled access. Users can search for items by entering exact specifications or general details, and the system will retrieve relevant pricing data from various sources. The interface should be minimalistic for fast performance, providing structured and tabulated results for easy comparison. The system should support different procurement methodologies, such as referencing past purchases or conducting market surveys, to enhance transparency and accuracy in cost estimation. By streamlining price benchmarking, the platform will improve procurement efficiency, reduce manual effort, and ensure cost reasonability in government purchases.

Problem Statement 20: A software application - Ground Water Level Predictor

Develop a web-based platform for groundwater level monitoring and analysis, providing users with real-time access to water level data collected from various agencies, including the Central Ground Water Board (CGWB). The system should feature a secure, user-friendly interface for visualizing groundwater trends using interactive maps and data dashboards. Users should be able to query specific locations and view historical groundwater levels, along with related factors such as rainfall, hydrogeology, land use, and population. The platform must support seamless integration with digital water level recorders (DWLRs) to display continuously updated data. Additionally, it should offer tools for data comparison, trend analysis, and reporting to support decision-making for water resource management. By centralizing groundwater data in a web-based system, the platform will improve accessibility, transparency, and efficiency in monitoring aquifer health across different regions.

Problem Statement 21: Procurement - Populate purchase history and price trends for specific item codes from SAP as well as online portals to assist buyer during RFP creation in Ariba

The procurement support web application aims to assist buyers during the Request for Proposal (RFP) creation process in Ariba by automatically populating purchase history and price trends for specific item codes. Traditional procurement processes often require manual data collection from SAP systems and online portals, leading to inefficiencies, errors, and delayed decision-making. This application will streamline the process by integrating with SAP and external sources to provide real-time, accurate pricing insights and historical purchase data. By offering comprehensive and up-to-date information, the system enables buyers to make informed decisions, optimize supplier negotiations, and enhance the efficiency of RFP creation.